Yanlong Zhao

Chapel Hill, NC | yanlongz@unc.com | School of Data and Information Sciences

Bio

I am a doctoral student in the School of Data and Information Sciences at the University of North Carolina at Chapel Hill. My research lies at the intersection of machine learning, computational oncology, and precision medicine. My long-term goal is to develop cutting-edge computational algorithms, AI models, and research platforms that integrate high-dimensional, multimodal cancer molecular data with functional genomics, proteomics, and clinical data to uncover novel cancer vulnerabilities and advance new therapeutic strategies. Through close collaboration with experimental scientists and oncologists, I aim to translate computational discoveries into actionable insights for cancer research and precision oncology.

Education

University of North Carolina at Chapel Hill, Ph.D. Student in Data and Information Science — Present

University of Rochester, M.S. in Electrical and Computer Engineering — Aug. 2023–May 2025

Shandong University, B.E. in Automation — Sep. 2019–Jun. 2023

Research Experience

University of North Carolina at Chapel Hill, Graduate Researcher – Chapel Hill, NC — Present
Advisor: Prof. Jun Li
Develop computational frameworks to identify novel biomarkers and potential therapeutic targets for chemotherapy-resistant small cell lung cancer (SCLC), and develop AI models to molecularly match cancer patients with relevant preclinical models, including cancer cell lines, organoids, and patient-derived xenograft (PDX) models.

Prof. Ren Wang@ TIML Lab (Co-advised by Prof. Can Chen), Research Assistant — Jul. 2024–Dec. 2025
Developed graph- and hypergraph-based machine learning methods that integrate network topology, molecular representations, and biomedical domain knowledge to enable data-driven biomedical discovery, with applications in drug repositioning and genome-scale metabolic model reconstruction.

Publications

A Multi-Way SMILES-Based Hypergraph Inference Network for Metabolic Model Reconstruction — 2026

Yanlong Zhao*, Yixiao Chen*, Yi Yu, Xiang Liu, Jiawen Du, Jun Wen, Quan Sun, Ren Wang, and Can Chen
Communications Biology, 9, Article 531. DOI:10.1038/s42003-026-09761-1

Dual-Route Embedding-Aware Graph Neural Networks for Drug Repositioning — 2025

Yanlong Zhao, Yixiao Chen, Jiawen Du, Jun Wen, Quan Sun, Ren Wang, and Can Chen
Briefings in Bioinformatics, 26(5), bbaf555. DOI:10.1093/bib/bbaf555

Awards & Honors

Lineberger Fellowship, UNC Lineberger Comprehensive Cancer Center — Oct. 2026 - Present

3rd Place Poster Prize, UNC Data Science Day — Sep. 2026
Deciphering Chemotherapy Resistance in Small Cell Lung Cancer through Integrative Proteomics.

Third-Class Academic Scholarship, Shandong University — Sep. 2022

Teaching Experience

DATA 110 @ University of North Carolina at Chapel Hill, Teaching Assistant — 2026 Fall

Industry Experience

DataOceanAI, Project Assistant – San Jose, CA — Nov. 2025–Jun. 2026

Support large-scale machine-learning data operations involving data collection, curation, quality control, and delivery.